Study Summary: Transformers

Overview

- Transformers process entire input sequences simultaneously using attention mechanisms, enabling fast training and scalable modeling.

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- The architecture is built from stacked layers of multi?head self?attention followed by feed?forward neural networks.

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- Two main variants exist: encoder?only models (e.g., BERT) for representation learning, and decoder?only models (e.g., GPT) for autoregressive text generation.

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Key Concepts

- Self?Attention Mechanism ? each token weighs the importance of all other tokens to capture contextual relationships regardless of distance.

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- Parallelization ? unlike RNNs, Transformers handle whole sequences at once, leading to faster training and better scalability.

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- Encoder?Decoder Structure ? the encoder produces rich semantic representations; the decoder uses these plus previously generated tokens to predict the next token.

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- Multi?Head Attention ? learns multiple patterns in parallel, providing diverse contextual embeddings.

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- Feed?Forward Neural Networks ? applied after each attention layer within a transformer block.

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- Autoregressive Output ? models generate one token at a time during decoding.

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- Encoder?Only Models (BERT?style) ? read text and produce contextual embeddings for both left and right context; used for classification, NER, QA.

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- Decoder?Only Models (GPT?style) ? generate text autoregressively.

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Algorithms & Formulas

- Attention lets each token decide which other tokens matter most, creating contextual embeddings.

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- Fine?tuning modifies the attention distribution for downstream tasks.

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- Pretrained transformers expose hidden states and attention weights that can be inspected or visualized.

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Practical Considerations

- Attention is powerful but computationally expensive because it requires pairwise interactions between all tokens in a sequence.

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- Fine?tuning changes what the model pays attention to, directly influencing downstream performance.

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- Encoder?only models are ideal for tasks needing contextual embeddings without generation; decoder?only models excel at text generation.

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